Gravitational Grille NRD/NRDD



NRD

Description
Rectangular aluminum grille for air overpressure with mobile blades (normally closed) .

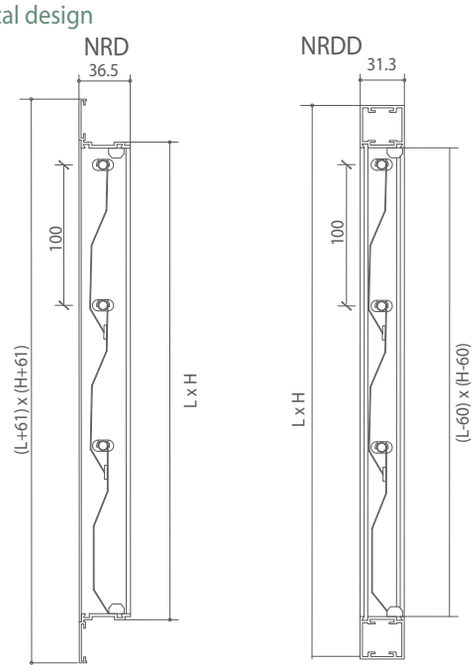
Application
Exhaust/ supply of the airflow

Characteristics
The standard grille is produced for air flow exhaust **NRD/NRDD**
Upon request, if the grid is used for introduction of air, the blades will be placed in reverse.
Plastic bearing sections to avoid friction.
Sealing gasket.
Central reinforcement for lengths over 500 mm.
Provided with mounting holes.

Accessories
Insulated/Non-insulated adapter **ADP IZ/N.**

Materials and finishing
Anodised frame with natural aluminum blades
Plastic bearing sections

Technical design



Assembly Methods
Countersinking holes on the visible frame of the grille (standard) **NRD**

Selection table NRD

Air velocity inside the grille Veff(m/s)	1	2	3	4	5	6	7
Pressure loss ΔPt (Pa)	10	18	25	30	36	44	52

Discharge area AK [m²]									
Height (mm)	Length (mm)								
	200	300	400	500	600	700	800	900	1000
200	0.03	0.05	0.07	0.08	0.10	0.12	0.14	0.15	0.17
300	0.05	0.08	0.10	0.13	0.16	0.19	0.21	0.24	0.27
400	0.07	0.10	0.14	0.18	0.22	0.25	0.29	0.33	0.37
500	0.08	0.13	0.18	0.23	0.27	0.32	0.37	0.42	0.46
600	0.10	0.16	0.22	0.27	0.33	0.39	0.45	0.50	0.56
700	0.12	0.19	0.25	0.32	0.39	0.46	0.52	0.59	0.66
800	0.14	0.21	0.29	0.37	0.45	0.52	0.60	0.68	0.76
900	0.15	0.24	0.33	0.42	0.50	0.59	0.68	0.77	0.85
1000	0.17	0.27	0.37	0.46	0.56	0.66	0.76	0.85	0.95

Selection table NRD - depending on air flow for velocity Veff = 3 m/s

Air flow [mc/h]									
Height (mm)	Length (mm)								
	200	300	400	500	600	700	800	900	1000
200	324	540	756	864	1080	1296	1512	1620	1836
300	540	864	1080	1404	1728	2052	2268	2592	2916
400	756	1080	1512	1944	2376	2700	3132	3564	3996
500	864	1404	1944	2484	2916	3456	3996	4536	4968
600	1080	1728	2376	2916	3564	4212	4860	5400	6048
700	1296	2052	2700	3456	4212	4968	5616	6372	7128
800	1512	2268	3132	3996	4860	5616	6480	7344	8208
900	1620	2592	3564	4536	5400	6372	7344	8316	9180
1000	1836	2916	3996	4968	6048	7128	8208	9180	10260

Annotation

* Veff[m/s] - Air velocity inside the grille
* ΔPt [pa] - Pressure loss
* Ak [m²] - Free surface

Note: Optimal velocity for net dimensions selection is between 2 and 4 m/s.
By applying the formula Ak[m²] = Air flow [mc/h] / 3600[s] / Veff [m/s] and using the selection table, the dimensions of the grille can be determined.

Order codes

Type + **ADP IZ/N** / **Dimensions** / **L X H**

NRD
NRDD

Insulated/Non-insulated adapter

L X H

200 x 200
300 x 300
400 x 400
500 x 500
600 x 600
700 x 700
800 x 800
900 x 900
1000 x 1000

Other dimensions on request